

1

Problem 1

Show that

$$\frac{dp_\mu}{d\tau} = \frac{q}{c} u^\nu F_{\mu\nu}$$

leads to

$$\frac{d\vec{p}}{dt} = q \vec{E} + q \frac{\vec{v}}{c} \times \vec{B}$$

and

$$\frac{d}{dt} \left(\frac{U}{c} \right) = q \frac{\vec{v}}{c} \cdot \vec{E}.$$

Problem 2

We can replace $d\tau = \frac{dt}{\gamma}$, and $U^0 = \gamma c$.

$$\begin{aligned} \gamma \frac{dp_\mu}{dt} &= \frac{q}{c} u^\nu F_{\mu\nu} \\ &= \frac{q}{c} (u^0 F_{\mu,0} + u^i F_{\mu,i}) \end{aligned}$$

Because $F_{\mu,\nu} = -F_{\nu,\mu}$, we have

$$\gamma \frac{dp_\mu}{dt} = -\frac{q}{c} \gamma (c F_\mu + u^i F_{i,\mu})$$

(note the γ because u is defined in proper time as $\frac{d}{d\tau} x_\mu$)

We know that $u^0 F_{0,\mu} = -c F_\mu$, and we can relabel $F_{0,i} = \vec{E}$. This gives us

$$\begin{aligned} \frac{dp_\mu}{dt} &= q \vec{E} - \frac{q}{c} u^i F_{i,\mu} \\ F_{\mu\nu} &= \begin{pmatrix} 0 & -E^1 & -E^2 & -E^3 \\ E_1 & 0 & B^3 & -B^2 \\ E^2 & -B^3 & 0 & B^1 \\ E^3 & B^2 & -B^1 & 0 \end{pmatrix} \end{aligned}$$

$$u = (c, \dot{x}, \dot{y}, \dot{z})$$

$$u^i F_{i,j} = \sum_{i=1}^3 v^i F_{i,j}$$

$$u^i F_{i,\mu} = (E_1, \dot{x}0, \dot{y}b^3, -\dot{z}b^2)i + (E_2, -\dot{x}b^3, \dot{y}0, \dot{z}b^1)j + (E_3, \dot{x}b^2, -\dot{y}b^1, 0\dot{z})k$$

writing out the cross product,

$$\vec{v} \times \vec{B} = \det \begin{pmatrix} \hat{i} & \hat{j} & \hat{k} \\ \dot{x} & \dot{y} & \dot{z} \\ b^1 & b^2 & b^3 \end{pmatrix}$$

$$= (\dot{y}b^3 - \dot{z}b^2)\hat{i} - (\dot{x}b^3 - \dot{z}b^1)\hat{j} + (\dot{x}b^2 - \dot{y}b^1)\hat{k}$$

We can see that the components with the magnetic field are the same vector, so $u^i F_{i,\mu} = \vec{v} \times \vec{B}$

Therefore, the whole expression is now

$$\frac{dp_\mu}{dt} = q \vec{E} - \frac{q}{c} u^i F_{i,\mu}$$

Therefore,

$$\frac{d\vec{p}}{dt} = q\vec{E} - \frac{q}{c}\vec{v} \times \vec{B}$$

$$\frac{dp_\mu}{d\tau} = \frac{q}{c} u^\nu F_{\mu\nu}$$

$$\frac{d}{dt} \left(\frac{U}{c} \right) = q \frac{\vec{v}}{c} \cdot \vec{E}.$$

We know that $F_{0,\nu} = \vec{E}$. Therefore, we can write out the sub expression

$$\begin{aligned} \frac{dp_0}{d\tau} &= \frac{q}{c} u^\nu \underbrace{F_{0\nu}}_{\vec{E}} \\ \gamma \frac{d}{dt} \left(\frac{U}{c} \right) &= \gamma \frac{q}{c} u^\nu E_\nu \\ \frac{d}{dt} \left(\frac{U}{c} \right) &= q(0 + \dot{x}' E_x + \dot{y}' E_y + \dot{z}' E_z) \\ \frac{d}{dt} \left(\frac{U}{c} \right) &= q\vec{v} \cdot \vec{E} \end{aligned}$$

Problem 2

Show that the Bianchi identity holds,

$$\epsilon^{\mu\nu\rho\sigma} \partial_\nu F_{\rho\sigma} = 0,$$

where

$$F_{\rho\sigma} = \partial_\rho A_\sigma - \partial_\sigma A_\rho.$$

Where are the other Maxwell equations

$$\vec{\nabla} \cdot \vec{B} = 0 \quad , \quad \vec{\nabla} \times \vec{E} = -\frac{1}{c} \frac{\partial \vec{B}}{\partial t} ? \quad (4.19)$$

They are the Bianchi identity

$$\epsilon^{\mu\nu\rho\sigma} \partial_\nu F_{\rho\sigma} = 0 \quad (4.20)$$

$$\begin{aligned} &= \epsilon^{\mu\nu\rho\sigma} \partial_\nu (\partial_\rho A_\sigma - \partial_\sigma A_\rho) \\ &= \epsilon^{\mu\nu\rho\sigma} \partial_\nu \partial_\rho A_\sigma - \epsilon^{\mu\nu\rho\sigma} \partial_\nu \partial_\sigma A_\rho \end{aligned}$$

Lets relabel $\nu = \rho$ and $\rho = \nu$.

$$\epsilon^{\mu\rho\nu\sigma} \partial_\rho \partial_\nu A_\sigma = \epsilon^{\mu\nu\rho\sigma} \partial_\nu \partial_\rho A_\sigma$$

These are the exact same,
we have just swapped the labeling
of a variable. We can swap the order
 $= -\epsilon^{\mu\rho\nu\sigma} \partial_\rho \partial_\nu A_\sigma$

The only solution to this is

$$\epsilon^{\mu\rho\nu\sigma} \partial_\rho \partial_\nu A_\sigma = 0$$

Therefore,

$$\begin{aligned} \epsilon^{\mu\nu\rho\sigma} \partial_\nu \partial_\rho A_\sigma - \epsilon^{\mu\nu\rho\sigma} \partial_\nu \partial_\sigma A_\rho &= 0 \\ \epsilon^{\mu\nu\rho\sigma} \partial_\nu F_{\rho\sigma} &= 0 \end{aligned}$$

Work from before recitation:

Hint: $\epsilon^{\mu,\nu,\rho,\sigma} \partial_\nu \partial_\rho A_\sigma = 0$

The totally antisymmetric tensor reduces this to only when μ, ν, ρ, σ are different indices (tacking on a negative sign sometimes). We can write out in summation notation and matrices to help me wrap my head around this

$$F_{\mu,\nu} = \begin{pmatrix} 0 & \partial_0 A_1 - \partial_1 A_0 & \partial_0 A_2 - \partial_2 A_0 & \partial_0 A_3 - \partial_3 A_0 \\ \partial_1 A_0 - \partial_0 A_1 & 0 & \partial_1 A_2 - \partial_2 A_1 & \partial_1 A_3 - \partial_3 A_1 \\ \partial_2 A_0 - \partial_0 A_2 & \partial_2 A_1 - \partial_1 A_2 & 0 & \partial_2 A_3 - \partial_3 A_2 \\ \partial_3 A_0 - \partial_0 A_3 & \partial_3 A_1 - \partial_1 A_3 & \partial_3 A_2 - \partial_2 A_3 & 0 \end{pmatrix}$$

$$\epsilon^{\mu\nu\rho\sigma} \partial_\nu (\partial_\rho A_\sigma - \partial_\sigma A_\rho) = \sum_{\nu=0}^3 \sum_{\rho=0}^3 \sum_{\sigma=0}^3 \epsilon^{\mu\nu\rho\sigma} \partial_\nu (\partial_\rho A_\sigma - \partial_\sigma A_\rho)$$

if $\mu = 0$, we have

$$\begin{aligned} \nu\rho\sigma &= \\ 123, 312, 231 &\quad \epsilon = +1 \\ 132, 321, 213 &\quad \epsilon = -1 \end{aligned}$$

I wrote these so that swapping the indices of the last two ρ and σ are paired.

For $\mu = 1, 2, 3$, we just replace that index in these six sequences with 0.

We can organize these to match so that the variables in the parenthesis cancel out, i.e. $\partial_\nu (\partial_2 A_3 - \partial_3 A_2)$ is the opposite of $\partial_\nu (\partial_3 A_2 - \partial_2 A_3)$. Because of ϵ , we are subtracting, and two negatives (one more from swapping the order in the parenthesis) makes them add.

Problem 3

Show that, under parity,

$$\vec{E}' = -\vec{E} \quad \vec{B}' = +\vec{B}$$

and under time reversal,

$$\vec{E}' = +\vec{E} \quad \vec{B}' = -\vec{B}.$$

Parity

EQ 4.21

$$j^{\mu'} = P_{\mu}^{\mu'} j^{\mu}$$

Eq 4.22

$$\begin{aligned} \partial^{\mu'} &= P_{\mu}^{\mu'} \partial^{\mu} \\ F^{\mu'\nu'} &= P_{\mu}^{\mu'} P_{\nu}^{\nu'} F^{\mu\nu} \\ \partial_{\mu} F^{\mu\nu} &= \frac{4\pi}{c} j^{\nu} \end{aligned}$$

We can transform this under parity

$$P_{\mu}^{\mu'} \partial_{\mu'} F^{\mu\nu} = \frac{4\pi}{c} P_{\nu}^{\nu'} j^{\nu'}$$

We can multiply both sides with the inverse

$$\begin{aligned}
 P_{\nu}^{\rho'} P_{\mu}^{\mu'} \partial_{\mu'} F^{\mu\nu} &= P_{\nu}^{\rho'} \frac{4\pi}{c} P_{\nu'}^{\nu} j^{\nu'} \\
 &= \delta_{\nu'}^{\rho'} \frac{4\pi}{c} J^{\nu'} \\
 \partial_{\mu'} (P_{\nu}^{\rho'} P_{\mu}^{\mu'} F^{\mu\nu}) &= \frac{4\pi}{c} j^{\rho'}
 \end{aligned}$$

This looks the same as before on the right side, so the internal part

$$P_{\nu}^{\nu'} P_{\mu}^{\mu'} F^{\mu\nu} = F^{\mu'\nu'}$$

Let's just directly apply parity to the components of F . The parity operator swaps the sign of the last three components in a four vector. Therefore, we swap the three columns going right and the three rows going down. The innermost items are hit twice, whereas the edges are only hit once. Therefore,

$$\begin{aligned}
 F^{\mu'\nu'} &= P_{\mu}^{\mu'} P_{\nu}^{\nu'} \begin{pmatrix} 0 & -E^1 & -E^2 & -E^3 \\ E_1 & 0 & B^3 & -B^2 \\ E_2 & -B^3 & 0 & B^1 \\ E_3 & B^2 & -B^1 & 0 \end{pmatrix} \\
 &= \begin{pmatrix} 0 & E^1 & E^2 & E^3 \\ -E_1 & 0 & B^3 & -B^2 \\ -E_2 & -B^3 & 0 & B^1 \\ -E_3 & B^2 & -B^1 & 0 \end{pmatrix}
 \end{aligned}$$

We can see that this has only swapped the components of the electric field, whereas the magnetic field has stayed the same. Therefore, under parity:

$$E' = -E, B' = B$$

Time reversal

Under time reversal transformation

$$\begin{aligned}
 j^{\mu'} &= -T_{\mu}^{\mu'} j^{\mu} \\
 \partial^{\mu'} &= +T_{\mu}^{\mu'} \partial^{\mu}
 \end{aligned}$$

Lets write out Maxwell's equations

$$\begin{aligned}
 \partial_{\mu} F^{\mu\nu} &= \frac{4\pi}{c} j^{\nu} \\
 \partial_{\mu'} F^{\mu'\nu'} &= \frac{4\pi}{c} j^{\nu'}
 \end{aligned}$$

The derivative only gets a T , but the J gets a $-$. Therefore, we have an additional minus sign at the beginning, but it looks the same as the parity bit. We can jump to:

$$\begin{aligned}
F^{\mu'\nu'} &= -T_{\mu}^{\mu'} T_{\nu}^{\nu'} \begin{pmatrix} 0 & -E^1 & -E^2 & -E^3 \\ E_1 & 0 & B^3 & -B^2 \\ E^2 & -B^3 & 0 & B^1 \\ E^3 & B^2 & -B^1 & 0 \end{pmatrix} \\
&= - \begin{pmatrix} 0 & E^1 & E^2 & E^3 \\ -E_1 & 0 & B^3 & -B^2 \\ -E^2 & -B^3 & 0 & B^1 \\ -E^3 & B^2 & -B^1 & 0 \end{pmatrix} \\
&= \begin{pmatrix} 0 & -E^1 & -E^2 & -E^3 \\ E_1 & 0 & -B^3 & B^2 \\ E^2 & B^3 & 0 & -B^1 \\ E^3 & -B^2 & B^1 & 0 \end{pmatrix}
\end{aligned}$$

We have swapped the E components twice, and the B components only once.

Therefore, under parity

$$\vec{E}' = \vec{E}, \vec{B}' = -\vec{B}$$

old bad work

Eq 4.8

$$\partial_{\mu} F^{\mu\nu} = \frac{-4\pi}{c} j^{\nu}$$

$$j^0 = \rho c$$

Under the parity coordinate transform, current does not change direction. Current is the source of the magnetic field.

$$\vec{\nabla} \times \vec{B} = \frac{4\pi}{c} \vec{J} + \frac{1}{c} \frac{d\vec{E}}{dt}$$

Under the coordinate transformation, the derivative does not exchange sign, so the sign of the term $\frac{d\vec{E}}{dt}$ will be the same in both coordinates. Therefore, transformed,

$$\vec{\nabla} \times \vec{B}' = \frac{4\pi}{c} P_{\mu}^{\mu'} j^{\mu} + \frac{1}{c} P_{\mu}^{\mu'} \frac{d\vec{E}}{dt}$$

$$\vec{\nabla} \cdot \vec{E} = \frac{4\pi}{c} j^0$$

Time reversal

$$\begin{aligned}
j^{\mu'} &= -T_{\mu}^{\mu'} j^{\mu} \\
\partial^{\mu'} &= +T_{\mu}^{\mu'} \partial^{\mu}
\end{aligned}$$

Problem 4

Starting from $F_{\mu\nu} = \partial_{\mu} A_{\nu} - \partial_{\nu} A_{\mu}$, show that $\vec{B} = \vec{\nabla} \times \vec{A}$.

I'm assuming that this is $\vec{\nabla} \times \vec{A}_i$ where $\vec{A}_i$ is the 1,2,3 components of $\vec{A}$ as a vector, because $\vec{B}$ has three components

First, we can recognize that in matrix form, $\partial_{\mu} A_{\nu} = (\partial_{\nu} A_{\mu})^T$.

We know that

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E^1 & -E^2 & -E^3 \\ E_1 & 0 & B^3 & -B^2 \\ E^2 & -B^3 & 0 & B^1 \\ E^3 & B^2 & -B^1 & 0 \end{pmatrix}$$

We can write out the components of $\vec{B}$ from this matrix

$$B^1 \text{ is } \mu = 3, \nu = 2$$

$$B^1 = \frac{\partial}{\partial x^2} A_3 - \frac{\partial}{\partial x^3} A_2$$

$$B^2 \text{ is } \mu = 3, \nu = 1$$

$$B^2 = \frac{\partial}{\partial x^3} A_1 - \frac{\partial}{\partial x^1} A_3$$

$$B^3 \text{ is } \mu = 2, \nu = 1$$

$$B^3 = \frac{\partial}{\partial x^1} A_2 - \frac{\partial}{\partial x^2} A_1$$

$$\begin{aligned} \vec{\nabla} \times \vec{A} &= \det \left(\begin{bmatrix} i & j & k \\ \frac{d}{dx^1} & \frac{d}{dx^2} & \frac{d}{dx^3} \\ A_1 & A_2 & A_3 \end{bmatrix} \right) \\ &= \left(\frac{d}{dx^2} A_3 - \frac{d}{dx^3} A_2 \right) i - \left(\frac{d}{dx^1} A_3 - \frac{d}{dx^3} A_1 \right) j + \left(\frac{d}{dx^1} A_2 - \frac{d}{dx^2} A_1 \right) k \end{aligned}$$

We can see that these components match - the i component has 2,3 - 3,2, the second component is 31-13, the last component is 1,2 - 2,1. QED.